

Internet of Things

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Motor

There are three different types of motors –

- DC motor
- Servo motor
- Stepper motor

DC Motor

A DC motor (Direct Current motor) is the most common type of motor. DC motors normally have just two leads, one positive and one negative. If you connect these two leads directly to a battery, the motor will rotate. If you switch the leads, the motor will rotate in the opposite direction.



Servo Motor

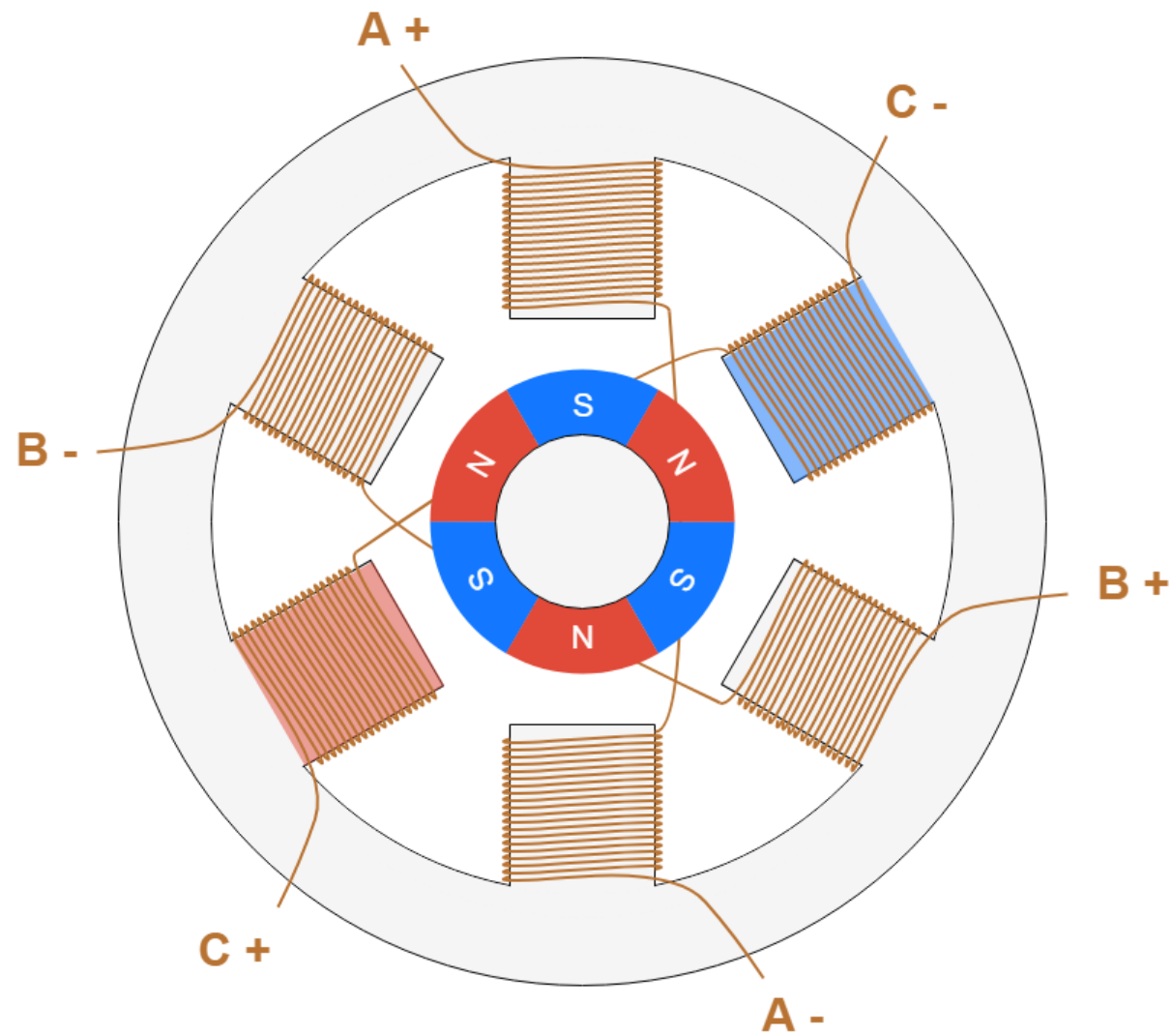
A **servomotor** (or **servo motor**) is a rotary actuator or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration. It consists of a suitable motor coupled to a sensor for position feedback. It also requires a relatively sophisticated controller, often a dedicated module designed specifically for use with servomotors.



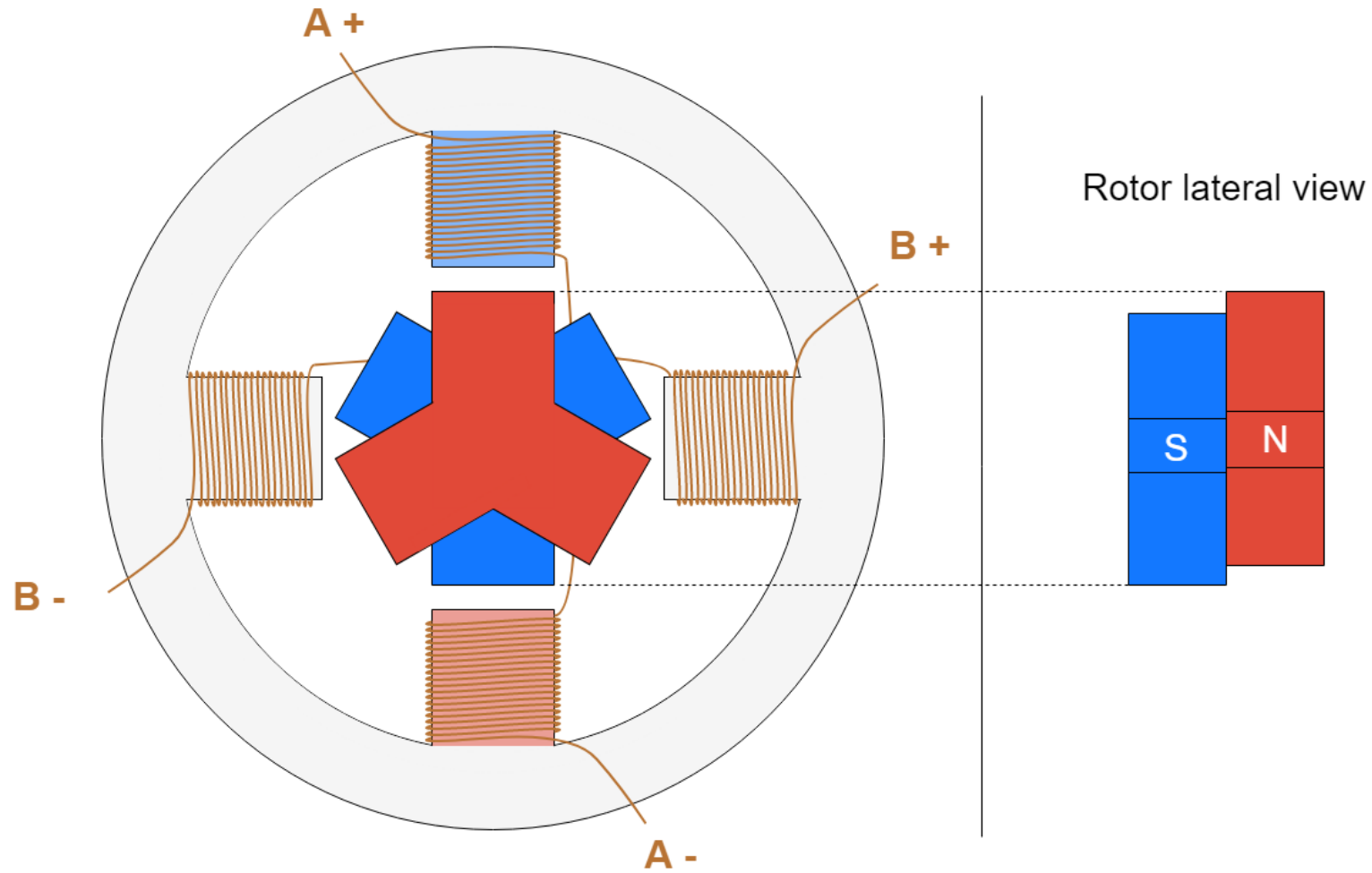
Stepper motor

A stepper motor, also known as a step motor or stepping motor, is a brushless DC electric motor that divides a full rotation into a number of equal steps. The motor's position can be commanded to move and hold at one of these steps without any position sensor for feedback (an open-loop controller), as long as the motor is correctly sized to the application with respect to torque and speed.

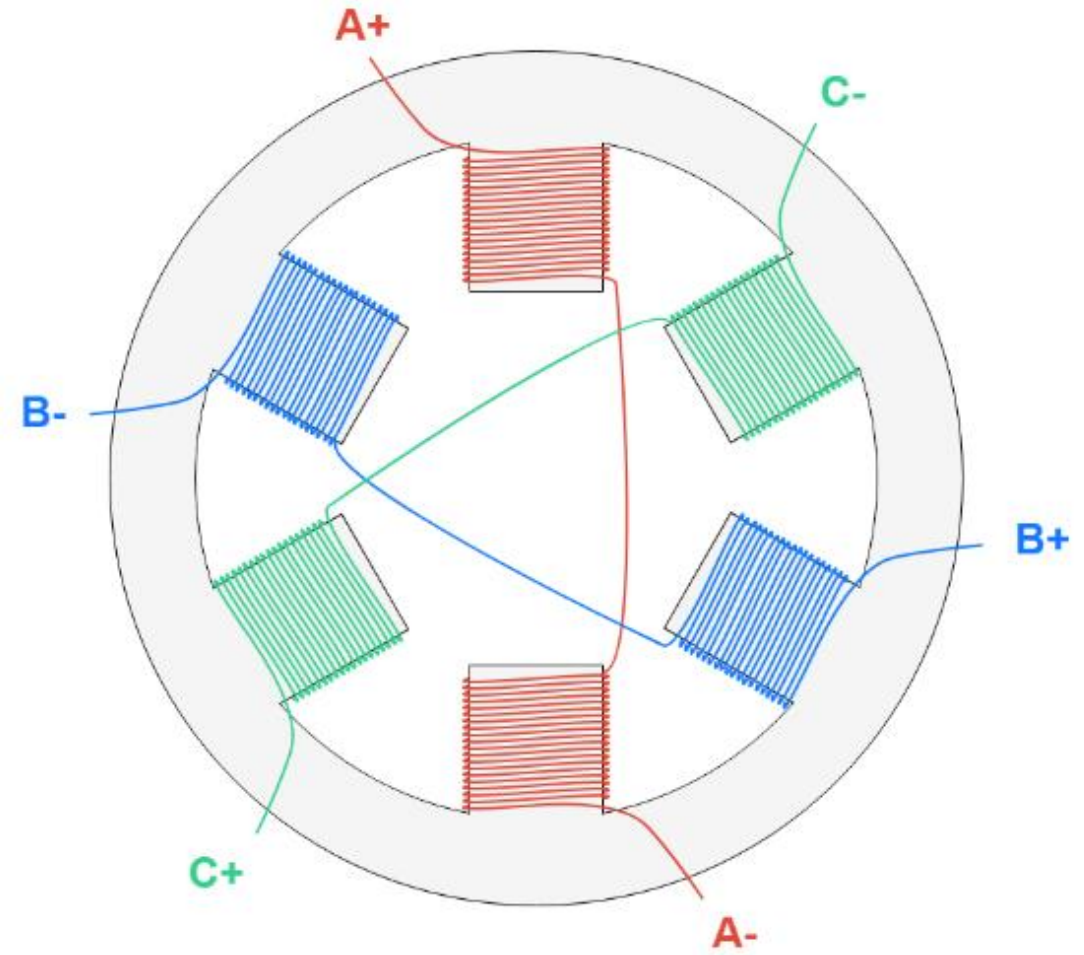
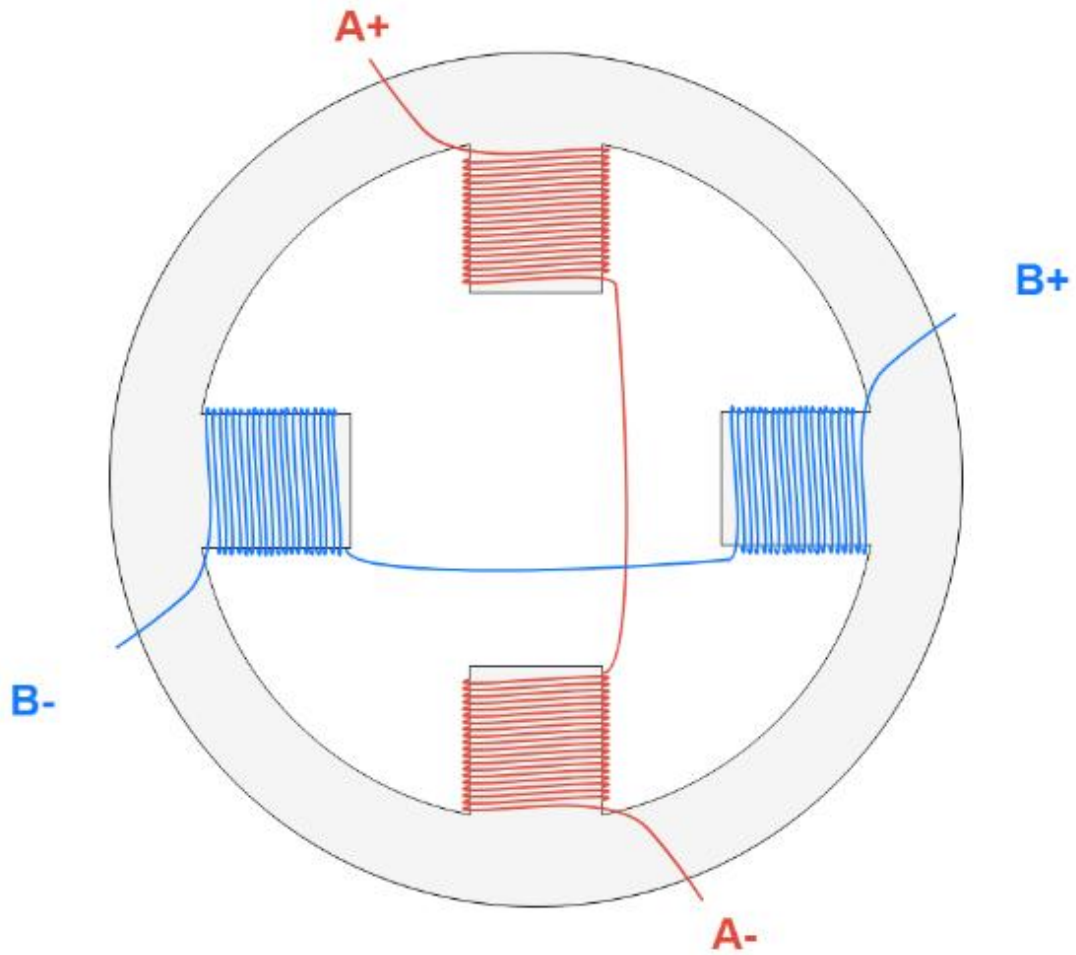




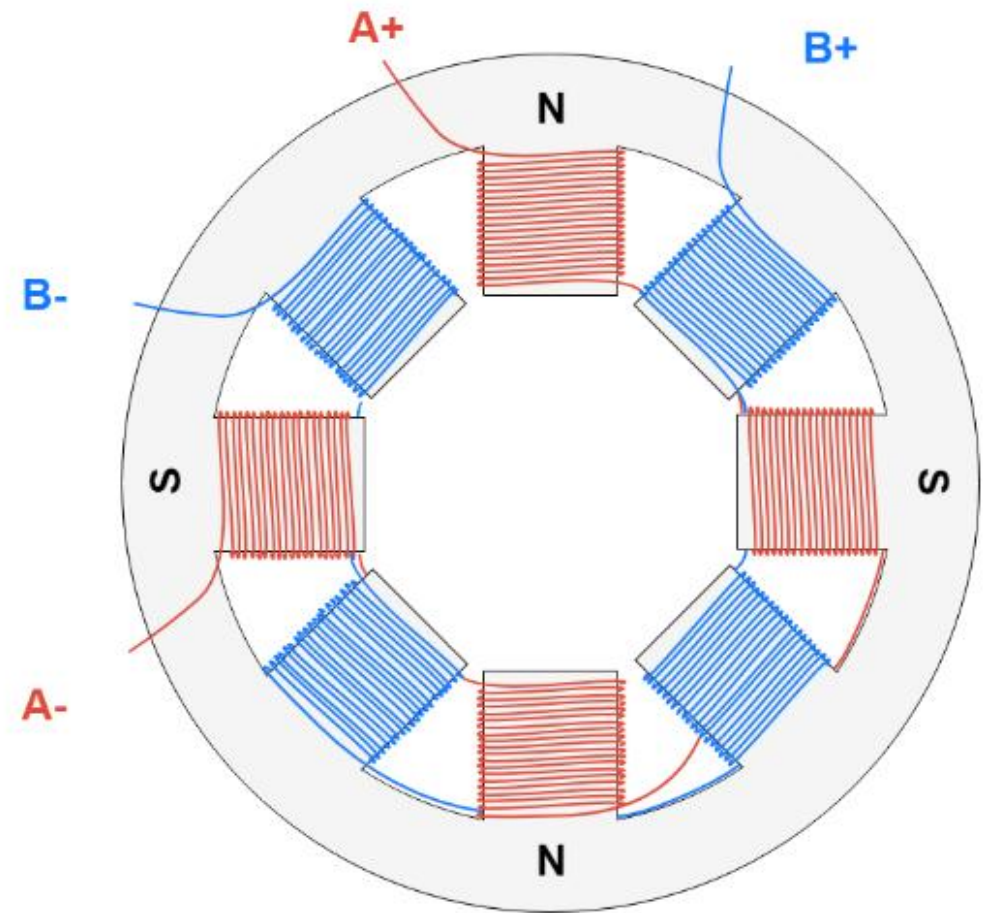
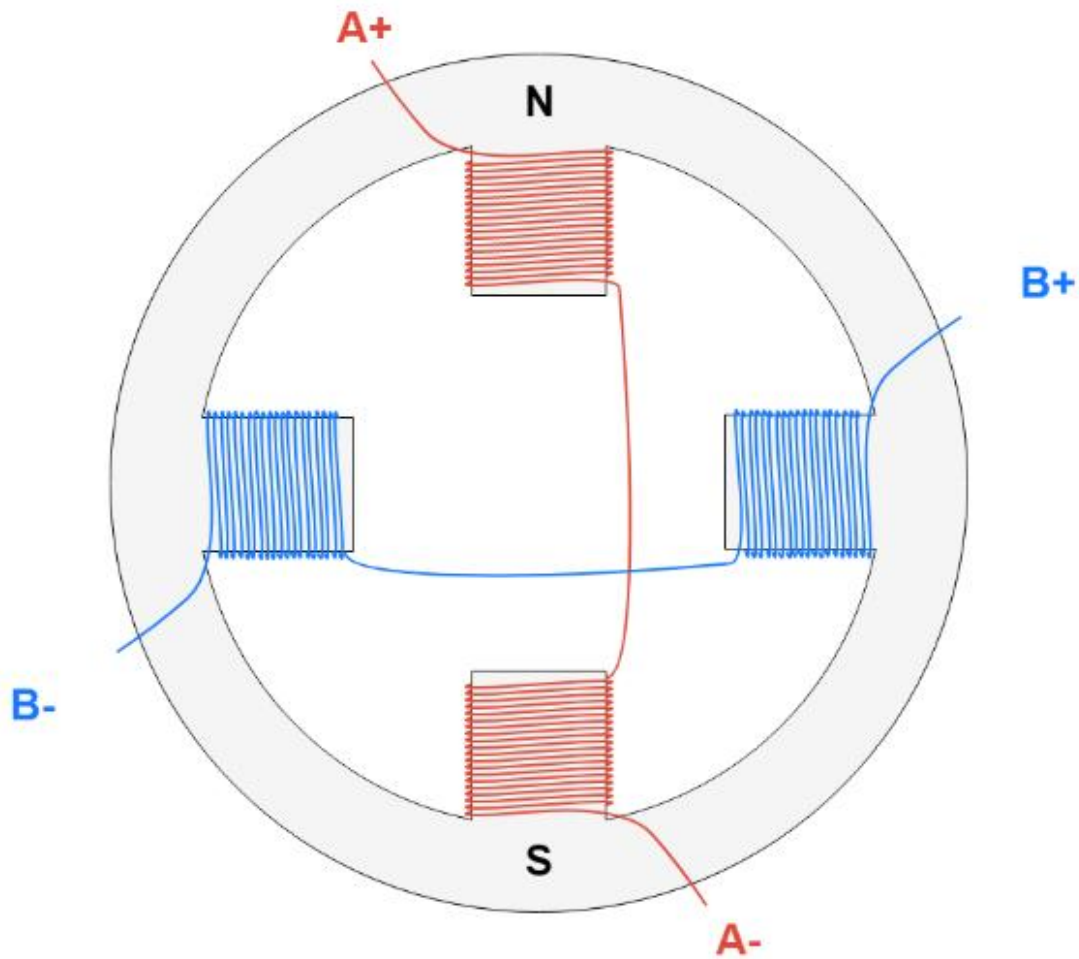
Permanent Magnet Stepper Motor



Hybrid Stepper Motor



Two-Phase Stator Winding (Left), Three-Phase Stator Winding (Right)



Two-Phase, Single-Pole Pair Stator (Left) and Two-Phase, Dipole Pair Stator (Right). The Letters Show the Magnetic Field Generated when Positive Voltage is Applied between A+ and A-.

Stepper Motor

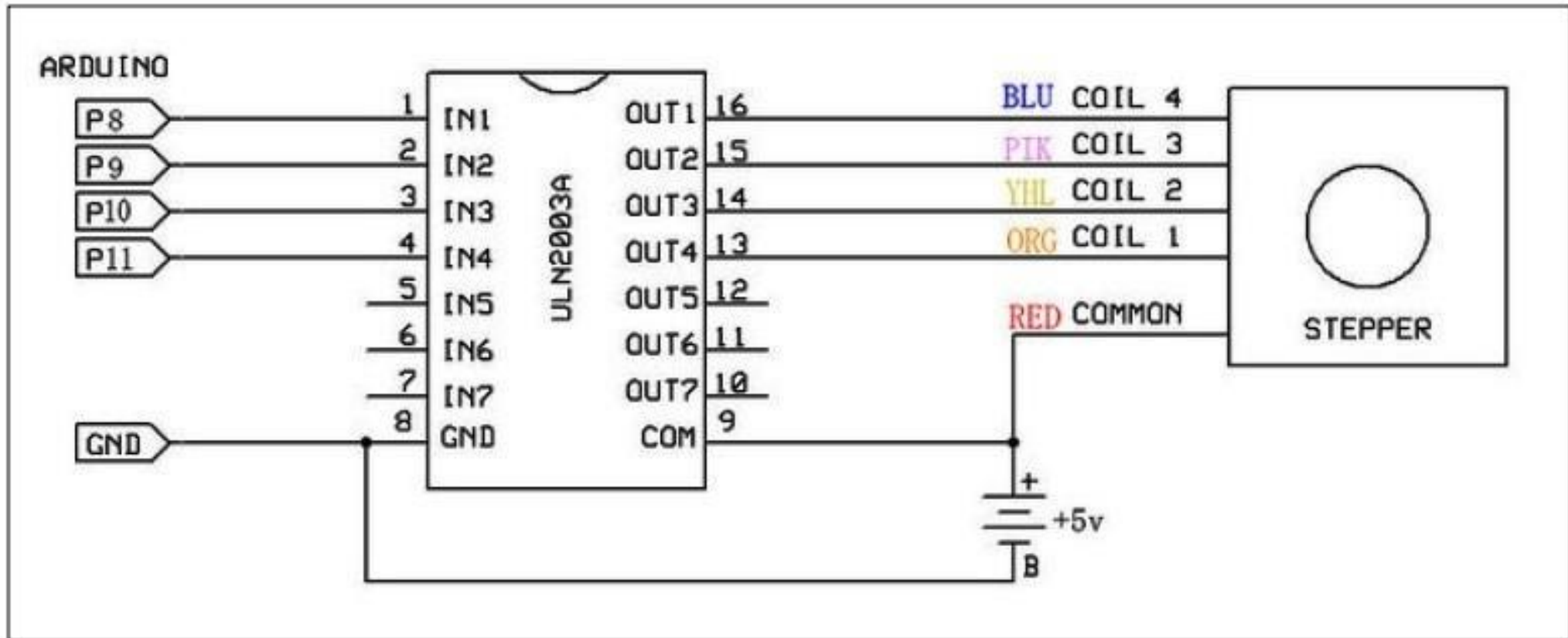
1. 28BYJ-48 5V Stepper Motor
2. ULN2003 Step Motor Driver Board
3. ULN2003A Step Moto Driver Board



28BYJ-48 5V Stepper Motor (Specs)

- Model 28BYJ-48 – 5V
- Rated voltage: 5VDC
- Number of phase: 4
- Speed variation ratio: 1/64
- Stride angle: $5.625^\circ / 64$
- Frequency: 100Hz
- DC resistance: $50\Omega \pm 7\%$ (25°C)
- Idle in-traction frequency: $> 600\text{Hz}$
- Idle out-traction frequency: $> 1000\text{Hz}$
- In-traction torque: $> 34.3\text{mN.m}$ (120Hz)
- Self-positioning torque: $> 34.3\text{mN.m}$
- Friction torque: 600-1200 gf.cm
- Pull in torque: 300 gf.cm
- Insulated resistance: $> 10\text{M}\Omega$ (500V)
- Insulated electricity power: 600VAC/1mA/1s
- Insulation grade: A
- Rise in temperature: $< 40\text{K}$ (120Hz)
- Noise: $< 35\text{dB}$ (120Hz, No load, 10cm)

Connection



Assignment

1. Interface with the Stepper Motor.
2. Connect the IR sensor with the Stepper motor and turn the motor to 90 degrees when an object is detected. Change the speed as well. (Parking Gate)
3. Connect the LDR with the motor and turn it according to different levels of luminosity. Change the Speed as well to simulate an automatic curtain system. (Automatic curtain system)

Thanks